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| 18 | D | <p>With the reduction in the volume of gas, the distance between the walls of the container is reduced, leading to an increase in the frequency of the collisions of the gas molecules with the walls. Hence the rate of change of momentum and hence force that the gas molecules experience increased. According to Newton's third law, the force and pressure acting on the walls increases.</p> <p>With constant temperature, the kinetic energy and speed of the gas molecules remains unchanged.</p> |
| 19 | B | $W = p(V_2 - V_1) = 1.3 \times 10^5 \times (3.6 - 1.3) \times 10^{-4}$                                                                                                                                                                                                                                                                                                                                                                                                                                     |